QUESTION 6: THREAT INTELLIGENCE PLATFORMS

DG Interns Hub Cybersecurity Internship – Week 5 Submission

Student Name: Gunjan **Track:** SOC Analyst Tier 1 **Topic:** VirusTotal, AbuseIPDB & AlienVault OTX

1. OVERVIEW OF THREAT INTELLIGENCE PLATFORMS

Threat Intelligence Platforms (TIPs) aggregate, correlate, and analyze threat data from worldwide security feeds. They provide L1 SOC analysts with context during alert triage, enabling quick determination of whether an IP, domain, URL, or file hash has been associated with known malicious activity.

2. KEY PLATFORMS BREAKDOWN

VirusTotal: A multi-engine aggregator that scans files, hashes, URLs, and IP addresses across 70+ antivirus engines and threat tools. It helps analysts identify known malware samples and suspicious external domains. **AbuseIPDB:** A crowd-sourced IP reputation repository focused on tracking IP addresses associated with web spam, port scanning, SSH/RDP brute-force attacks, and DDoS attempts. **AlienVault OTX (Open Threat Exchange):** A open threat intelligence community that allows security researchers to share bundled threat data called *Pulses*. These contain grouped indicators, threat actor profiles, and YARA rules.

3. DETAILED COMPARISON MATRIX

Platform	Primary Focus	Supported Indicators	Common Use Case	How L1 Analyst Uses It
VirusTotal	Multi-engine antivirus & URL reputation analysis.	File Hashes (MD5, SHA256), URLs, IPs, Domains.	Analyzing email attachments or command-line executable hashes.	Pastes a file hash or URL into VirusTotal to see if 50+ vendors flag it as Trojan/Ransomware.
AbuseIPDB	Crowd-sourced IP abuse confidence scoring.	IPv4 & IPv6 Addresses.	Investigating inbound brute-force or port scan alerts.	Queries source IP address to check Abuse Confidence Score (%) and recent report logs.
AlienVault OTX	Crowd-sourced threat dossiers & campaign tracking.	IPs, Hashes, Domains, YARA, CVEs.	Attributing indicators to known threat groups (APTs).	Checks if an IP or hash belongs to an active Threat Pulse linked to broader campaigns.

4. WHY ANALYSTS SHOULD NOT BLINDLY TRUST TI RESULTS

Critical Insight: The Caveats of Threat Intelligence

Threat intel feeds provide valuable evidence, but they must be interpreted alongside internal organizational context:

Zero-Day & Newly Compiled Malware: A brand new custom payload will show 0/72 detections on VirusTotal because antivirus vendors have not created signatures for it yet. **Shared Cloud & CDN Infrastructure:** Malicious actors often use legitimate hosting (AWS, Cloudflare, Azure). Blocking an IP based solely on old TI reports might break legitimate cloud services. **Internal Context Overrules Public Intel:** Threat intel platforms have zero visibility into your internal corporate network. An internal IP performing network scans won't appear on AbuseIPDB, but it could still be an active internal compromise.

5. TERMINOLOGY & REFERENCES

IOC (Indicator of Compromise): Static evidence of malicious activity (e.g., file hash, malicious IP).
Abuse Confidence Score: AbuseIPDB metric (0–100%) indicating how frequently an IP address is reported for attack activity.

References:

• VirusTotal Documentation: **https://www.virustotal.com/**
• AbuseIPDB API & Data Guides: **https://www.abuseipdb.com/**
• AlienVault OTX Community Pulse Feed: **https://otx.alienvault.com/**